



DATA GENERATION

MatCalc steel database
nine Fe-base alloy systems

phase fractions



DATA PARTITIONING

cluster-stratified 64 / 16 / 20
KMeans $k = 6$, three seeds

training / validation / test



SURROGATE TRAINING

simplex-constrained MLP + gated head
ridge, k-NN, XGBoost, forest

$\sum f_i = 1$

fitted surrogates



GENERALIZATION TESTS

interpolation, band holdout,
extrapolation; ideal-form control

phase fractions at (x, T)

HIGH-THROUGHPUT SCREENING

experimental anchor and
two screening campaigns

held-out rows